$\nabla = \frac{\partial}{\partial x}\vec{x_0} + \frac{\partial}{\partial y}\vec{y_0} + \frac{\partial}{\partial z}\vec{z_0}$

$\nabla F = \frac{\partial F}{\partial x}\vec{x_0} + \frac{\partial F}{\partial y}\vec{y_0} + \frac{\partial F}{\partial z}\vec{z_0}$

$\nabla\cdot\vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$

$\nabla\times\vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\vec{x_0} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\vec{y_0} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\vec{z_0}$

对于 $\vec{A}(\rho,\varphi,z)$：$\nabla\cdot\vec{A} = \frac{1}{\rho}\frac{\partial(\rho A_\rho)}{\partial\rho} + \frac{1}{\rho}\frac{\partial A_\varphi}{\partial\varphi} + \frac{\partial A_z}{\partial z}$

$\nabla\times\vec{A} = \frac{1}{\rho}\begin{vmatrix} \vec{\rho_0} & \rho\vec{\varphi_0} & \vec{z_0} \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\varphi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\varphi & A_z \end{vmatrix}$

对于 $\vec{A}(r,\theta,\varphi)$：$\nabla\cdot\vec{A} = \frac{1}{r^2}\frac{\partial(r^2A_r)}{\partial r} + \frac{1}{r\sin\theta}\left[\frac{\partial(A_\theta\sin\theta)}{\partial\theta} + \frac{\partial A_\varphi}{\partial\varphi}\right]$

$\nabla\times\vec{A} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \vec{r_0} & r\vec{\theta_0} & r\sin\theta\,\vec{\varphi_0} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\varphi} \\ A_r & rA_\theta & r\sin\theta A_\varphi \end{vmatrix}$

$\text{grad}\,\Phi = \nabla\Phi$

$\text{div}\vec{A} = \nabla\cdot\vec{A} \xrightarrow{=0}$ 无源场

$\text{curl}\,\vec{A} = \nabla\times\vec{A} \xrightarrow{=0}$ 无旋场

无源场 & 无旋场 → 调和场

极化

$\vec{E} = (\vec{x}E_x + \vec{y}E_y)e^{-jkz}$，$\frac{E_y}{E_x} = Ae^{j\varphi}$

① $\varphi = 0 / \pi \to$ LP

② $\varphi = \pm\frac{\pi}{2}$，$A=1 \to$ CP：$\varphi=\frac{\pi}{2} \to$ LHCP；$\varphi=-\frac{\pi}{2} \to$ RHCP.

③ 上半平面 → LH 椭圆~；下半平面 → RH ~.

平面波

① 无源、简单介质中（无耗），Maxwell：
$\begin{cases} \nabla\times\vec{E} = -j\omega\mu\vec{H} \\ \nabla\times\vec{H} = j\omega\varepsilon\vec{E} \\ \nabla\cdot\vec{E} = 0 \\ \nabla\cdot\vec{H} = 0 \end{cases}$

· 传播常数 $k = \omega\sqrt{\mu\varepsilon}$

· 波矢：$\vec{k} = k_x\vec{x_0} + k_y\vec{y_0} + k_z\vec{z_0}$

· 传播方向 $\vec{K} = \frac{\vec{k}}{|\vec{k}|}$，波长 $\lambda = \frac{2\pi}{|\vec{k}|}$

· $Z = \frac{1}{Y} = \frac{k}{\omega\varepsilon} = \sqrt{\frac{\mu}{\varepsilon}} = \eta$

② 有耗：$\tilde{\varepsilon} = \varepsilon - j\frac{\sigma}{\omega}$，$\vec{k} = k_r - jk_i$

▲ σ 很小的介质：$k_i = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$，$k_r = \omega\sqrt{\mu\varepsilon}$.

σ 很大的：$k \approx \sqrt{\frac{\omega\mu\sigma}{2}}\cdot(1-j)$

↳ · 穿透深度 $d_p = \sqrt{\frac{2}{\omega\mu\sigma}} \stackrel{def}{=} \delta$ 趋肤深度

∴ 对于良导体，电磁场主要集中在 δ 厚度的薄层内.

则 · $\vec{E} = \vec{x_0}E_0e^{-k_iz}\cos(\omega t - k_rz)$

(or $\vec{E} = \vec{x_0}E_0e^{-k_iz}e^{-k_rz}$)

$\vec{H} = \vec{y_0}\frac{E_0}{|\eta|}e^{-k_iz}e^{-jk_rz}e^{-j\varphi}$　　$\vec{E}(r) = |\eta|\vec{H}(r)\times\vec{K}$

·· 波沿 +z 传播的 $v = \frac{\omega}{k_r}$

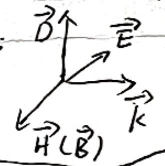
··· $d_p = \frac{1}{k_i}$

③ 电各向异性介质

$\vec{D} = \tilde{\varepsilon}\cdot\vec{E}$

$\vec{B} = \tilde{\mu}\cdot\vec{H}$

now is：



$(\vec{D}, \vec{E}, \vec{k}, \vec{S}$ 共面$)$

· 群速 v_g：

① 非色散介质（v_p 与 ω 无关）$v_g = v_p$

② 否则，$v_g = \frac{v_p}{1 - \frac{\omega}{v_p}\left(\frac{dv_p}{d\omega}\right)}$

正常色散 ← $\frac{dv_p}{d\omega} < 0$；$\frac{dv_p}{d\omega} > 0 \to$ 非正常色散

平行双导线

无损耗时：$U = U^ie^{-jkz} + U^re^{jkz}$

$I = \frac{U^ie^{-jkz} - U^re^{jkz}}{Z_c}$

$\left(\eta = \sqrt{\frac{\mu}{\varepsilon}} = |\eta|e^{j\varphi}\right)$

有损耗时：$U = U^ie^{-jk_rz}e^{-k_iz} + U^re^{jk_rz}e^{k_iz}$

$(k = k_r - jk_i)$

$I = \frac{1}{Z_c}\left(U^ie^{-jk_rz}e^{-k_iz} - U^re^{jk_rz}e^{k_iz}\right)$

· 反射系数 $\Gamma_u(z) = \frac{U^re^{jkz}}{U^ie^{-jkz}} = \frac{U^r}{U^i}e^{2jkz} = \Gamma_u(0)e^{2jkz} = \frac{Z(z)-Z_c}{Z(z)+Z_c}$

↳ $Z(z) = \frac{U(z)}{I(z)} = Z_c\frac{1+\Gamma_u(z)}{1-\Gamma_u(z)}$

· $Z_{in} = Z_c\frac{Z_L + jZ_c\tan kl}{Z_c + jZ_L\tan kl}$

· $\Gamma_u(0) = \frac{Z_L - Z_c}{Z_L + Z_c} = |\Gamma_u(0)|e^{j\varphi(0)}$，
$\begin{cases} |\Gamma_u(0)| = \left|\frac{Z_L - Z_c}{Z_L + Z_c}\right| \\ \varphi(0) = \arctan\left(\frac{2X_LZ_c}{R_L^2 + X_L^2 - Z_c^2}\right) \end{cases}$

(记 $Z_L = R_L + jX_L$)

$\begin{cases} U_{max} = [1 + |\Gamma_u(0)|] \times 电压幅度 \\ U_{min} = [1 - |\Gamma_u(0)|] \times 电压幅度 \end{cases}$

· $d_{max1} = \frac{\varphi(0)\lambda}{4\pi}$，$d_{min1} = d_{max1} + \frac{\lambda}{4}$

· VSWR：$\rho = \frac{U_{max}}{U_{min}} = \frac{1+|\Gamma_u|}{1-|\Gamma_u|}$

边界条件

切向：
$\vec{n_0}\times(\vec{H_1} - \vec{H_2}) = \vec{J_S}$　表面电流
$\vec{n_0}\times(\vec{E_1} - \vec{E_2}) = 0$

法向：
$\vec{n_0}\cdot(\vec{D_1} - \vec{D_2}) = \rho_S$　对于 σ→∞
$\vec{n_0}\cdot(\vec{B_1} - \vec{B_2}) = 0$　表面电荷密度

[对于 σ 有限的介质，切向电场和磁场均连续

① $\theta_i = \theta_r$

② $n_1\sin\theta_1 = n_2\sin\theta_2$　$(n = \sqrt{\varepsilon_r})$

Critical Angle：① 光密 → 疏 ② $\theta_i > \theta_c$（全反射）

$\sin\theta_c = \frac{n_2}{n_1} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$

Brewster Angle：① 入射波是 TM 波 ② $\theta_i = \theta_b$（入射波全折射）

$\tan\theta_b = \frac{n_2}{n_1} = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$　（且有 $\theta_b + \theta_t = \frac{\pi}{2}$）

传输功率 $P(z) = \frac{1}{2}\text{Re}[U(z)\cdot I^*(z)]$　$\left(\frac{P^r}{P^i} = |\Gamma_u|^2\right)$

对于无耗传输线：$P = P^i - P^r = \frac{1}{2}\frac{|U^i|^2}{Z_c} - \frac{1}{2}\frac{|U^i|^2}{Z_c}|\Gamma_u|^2$.

$P = \frac{1}{2}\frac{|U_{max}||U_{min}|}{Z_c} = \frac{1}{2}\frac{|U_{max}|^2}{Z_c\rho}$.

$\rho = \nabla\cdot\vec{D} = \varepsilon\cdot\nabla\vec{E} = \varepsilon\left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\right)$

① 吸收介质界面：

无耗 ε_{r1}，k_{z1}, Z_1；有耗介质 $\varepsilon_{r2} = \varepsilon_{r2}' - j\varepsilon_{r2}''$，$k_{z2}, Z_2$

[非均匀平面波]

$k_{z1} = k_1\cos\theta = k_0\cos\theta$

$k_{z2} = \sqrt{k^2 - k_{x1}^2} = \sqrt{k^2 - (k_1\sin\theta)^2} = k_0\sqrt{\tilde{\varepsilon}_{r2} - \sin^2\theta}$

$Z_1 = \begin{cases} \frac{\omega\mu}{k_{z1}} \\ \frac{k_{z1}}{\omega\varepsilon} \end{cases}$　$Z_2 \begin{cases} \frac{\omega\mu}{k_{z2}} \\ \frac{k_{z2}}{\omega\varepsilon} \end{cases}$　$\Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1}$　$T = \begin{cases} 1+\Gamma \\ 1-\Gamma \end{cases}$

瞬时坡印廷功率流 $\vec{S}(\vec{r},t) = \vec{E}(\vec{r},t)\times\vec{H}(\vec{r},t)$

↳ 平均功率密度 $\langle\vec{S}(t)\rangle = \frac{1}{2}\text{Re}[\vec{E}(r)\times\vec{H}^*(r)] = \frac{1}{2}\sqrt{\frac{\varepsilon}{\mu}}|E_0|^2\vec{z_0}$

复数 ~：$\vec{S}(\vec{r}) = \vec{E}(\vec{r})\times\vec{H}^*(\vec{r})$.

$w_e(z,t) = \frac{1}{2}\varepsilon[\vec{E}(z,t)]^2$

$w_m(z,t) = \frac{1}{2}\mu[\vec{H}(z,t)]^2$

$w_e = \frac{1}{2}\varepsilon|\vec{E}(\vec{r})|^2$

$w_m = \frac{1}{2}\mu|\vec{H}(\vec{r})|^2$

② 介质-导体界面

k_{z1}, Z_1 — Z_m　$Z_m = R(1+j) = \sqrt{\frac{\omega\mu_0}{2\sigma}}\cdot(1+j)$，$\Gamma = \frac{Z_m - Z_1}{Z_m + Z_1}$

若从完纯导体表面，则会全反射 $(\Gamma \to -1)$

TE（垂直极化波）　　　　　　　　TM（平行极化波）

入：$\vec{E}^i = \vec{y}_0 a e^{-jk_x^i x - jk_z^i z}$　　　$\vec{H}^i = \vec{y}_0 a e^{-jk_x^i x - jk_z^i z}$

$\vec{H}^i = H_x \vec{x}_0 + H_z \vec{z}_0 \quad (H_y = 0)$　　　$\vec{E}^i = E_x \vec{x}_0 + E_z \vec{z}_0 \quad (E_y = 0)$

$\vec{k} = k_x \vec{x}_0 + k_z \vec{z}_0 \quad (k_y = 0)$　　　$\vec{k} = k_x \vec{x}_0 + k_z \vec{z}_0 \quad (k_y = 0)$

反：$\vec{E}^r = \vec{y}_0 b e^{-jk_x^r x + jk_z^r z}$　　　$\vec{H}^r = b e^{-jk_x^r x + jk_z^r z}$

$\begin{cases} H_x^r = \frac{1}{j\omega\mu}\left(-\frac{\partial E_y^r}{\partial z}\right) \\ H_z^r = \frac{1}{j\omega\mu}\left(-\frac{\partial E_y^r}{\partial x}\right) \end{cases}$　　　$\begin{cases} E_x^r = \frac{1}{j\omega\varepsilon_1}\left(-\frac{\partial H_y^r}{\partial z}\right) \\ E_z^r = \frac{1}{j\omega\varepsilon_1}\left(\frac{\partial H_y^r}{\partial x}\right) \end{cases}$

折：$\vec{E}^t = \vec{y}_0 c e^{-jk_x^t x - jk_z^t z}$　　　$\vec{H}^t = c e^{-jk_x^t x - jk_z^t z}$

$\begin{cases} H_x^t = \frac{1}{j\omega\mu}\left(\frac{\partial E_y^t}{\partial z}\right) \\ H_z^t = \frac{1}{j\omega\mu}\left(-\frac{\partial E_y^t}{\partial x}\right) \end{cases}$　　　$\begin{cases} E_x^t = \frac{1}{j\omega\varepsilon_2}\left(-\frac{\partial H_y^t}{\partial z}\right) \\ E_z^t = \frac{1}{j\omega\varepsilon_2}\left(\frac{\partial H_y^t}{\partial x}\right) \end{cases}$

$Z_1 = \frac{\omega\mu_0}{k_{z_1}}$　　　$Z_1 = \frac{k_{z_1}}{\omega\varepsilon}$

· 多层介质平板：　Ⅰ　Ⅱ　Ⅲ

$z=0$　$z=d$　z

Z_1　Z_2　Z_3

$k_1, k_2, k_3 = \omega\sqrt{\mu\varepsilon}$

$k_{x1}, k_{x2}, k_{x3} = k_1 \sin\theta_i$

$k_{z_1} = \sqrt{k_1^2 - k_{x1}^2}$，$k_{z_2} = \sqrt{k_2^2 - k_{x2}^2} = \sqrt{k_2^2 - k_{x1}^2}$，$k_{z_3}$

$Z_1 = \frac{\omega\mu_0}{k_{z_1}}$，$Z_2$，$Z_3$.

$Z_{in}(0^-) = Z_2 \frac{Z_3 + jZ_2\tan(k_{z_2}d)}{Z_2 + jZ_3\tan(k_{z_2}d)}$

$\Gamma(0^-) = \frac{Z_{in}(0^-) - Z_1}{Z_{in}(0^-) + Z_1}$

$z=0$ 交界面电压、电流为：$U(0) = [1+\Gamma(0^-)]U_1^i$，$I(0) = [1-\Gamma(0^-)]\frac{U_1^i}{Z_1}$

区域Ⅰ：$U_{1max} = (1+|\Gamma(0^-)|)U_1^i$

$U_{1min} = (1-|\Gamma(0^-)|)U_1^i$

$d_{min1} = \frac{\varphi(0)}{4\pi}\lambda + \frac{\lambda}{4}$　亦以°为单位　换成720°计算

Ⅱ：$\Gamma(d^-) = \frac{Z_3 - Z_2}{Z_3 + Z_2}$

$\Gamma(0^+) = \Gamma(d^-) \cdot e^{-j2k_{z_2}d}$

$U_2^i = \frac{1+\Gamma(0^-)}{1+\Gamma(0^+)} U_1^i$

$U_2(d^-) = [1+\Gamma(d^-)]U_2^i$，$I_2(d) = [1-\Gamma(d^-)]\frac{U_2^i}{Z_3}$

$U_{2max} = (1+|\Gamma(d^-)|)|U_2^i|$，$I_{2max} = \frac{U_{2max}}{Z_2}$

$U_{2min} = (1-|\Gamma(d^-)|)|U_2^i|$，$I_{2min} = \frac{U_{2min}}{Z_2}$

Ⅲ：（行波）$U_3(d^+) = U_2(d^-)$，$I_3(d^+) = I_2(d^-)$

电偶极子 $\vec{p}$　$\vec{p} = q\vec{d}$　$\vec{d}$：$\ominus \xrightarrow{} \oplus$，$-q$　$+q$

磁偶极子 $\vec{m} = IS\vec{z}_0$　　$\vec{z}_0$

（圆图）

① Γ：连接OA交 $|\Gamma|=1$ 的圆于点C（i.e. 最外圆）

以O点为圆心

则 $|\Gamma_A| = \frac{\overline{OA}}{\overline{OC}}$

② $\overline{OA}$ 与右实轴夹角即为 Γ 的相角 φ

③ 等 $|\Gamma_A|$ 圆：以O点为圆心，|OA| 为 r 的

等 $|\Gamma_A|$ 圆与右实轴的交点 读出数即为

④ 点A 顺时针沿等 $|\Gamma_A|$ 圆转至左半径，电长度即为 $\frac{d_{min}}{\lambda}$

⑤ 圆图 左端点：终端短路（$Z_L=0$）

右　　　　开

阻抗匹配

① $\frac{\lambda}{4}$：要求负载 Z_L 为纯电阻 R_L.

一段 $\frac{\lambda}{4}$ 的传输线，其 $Z_{c_1} = \sqrt{R_L Z_c}$

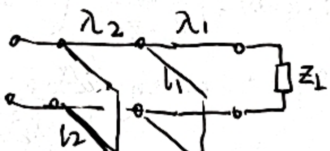
② 并联支路可变电纳匹配器：

单、从 Z_L 的导纳点A 沿等 $|\Gamma_A|$ 圆 顺时针 旋转，与 $g=1$ 的等 g 圆 交 2个点 B、C，

按此电长度 调节 l_B / l_C.

从导纳圆图短路点（右端点）沿 $|\Gamma|=1$ 的圆 顺时针转至以上的电长度

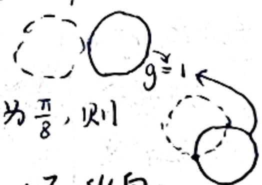
双、归一化 $y_L \xrightarrow[\text{顺时针 } \lambda_1]{\text{沿等}\Gamma\text{圆}} y_A \xrightarrow{\text{沿等g圆转}}$ 与虚线圆交于 B、C点



取决于 λ_2，

若为 $\frac{\lambda}{4}$，则

若为 $\frac{\pi}{8}$，则



顺～ λ_2　沿等 Γ 圆

等"g=1"圆上 D.

沿等g圆

"1+j0" 点、F

平面波传输线模型（z—纵向　t—横向）

TE（横电）模　　　TM（横磁）模

$\vec{E}'(\vec{r}) = \vec{E}_t'(\vec{r})$　　　$\vec{E}''(\vec{r}) = \vec{E}_t''(\vec{r}) + E_z''(\vec{r})\vec{z}_0$

$E_z' = 0$　　　$\vec{H}''(\vec{r}) = \vec{H}_t''(\vec{r})$

$\vec{H}'(\vec{r}) = \vec{H}_t'(\vec{r}) + H_z'\vec{z}_0$　　　$H_z''(\vec{r}) = 0$

$Z' = \frac{\omega\mu}{k_z'} = \frac{1}{Y'}$　　　$Z'' = \frac{1}{Y''} = \frac{k_z''}{\omega\varepsilon}$

法拉第电磁感应 Law：变化的 $B \xrightarrow{generate} E$.

位移电流假说：变化的 $E \longrightarrow B$.

Maxwell 提出了位移电流的概念.

当介质电导率很大时，传导电流 ≫ 位移电流（通常可忽略不计）

理想介质中的均匀平面波 is 非色散波

导电媒质　～　　～ 色散波